



CBCS SCHEME

BCS402

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Microcontroller

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain the purpose of various fields of current program status register with a neat diagram.	05	L2	CO1
	b.	Explain the ARM design philosophy.	06	L2	CO1
	c.	Explain the core extensions of ARM processor with neat block diagram.	09	L2	CO1
OR					
Q.2	a.	Explain Embedded systems hardware with a neat block diagram.	06	L2	CO1
	b.	What is pipelines in ARM? Illustrate with an example the pipeline stage of ARM 9 and ARM 10.	09	L2	CO1
	c.	Describe the RISC design philosophy with 4 design rules.	05	L2	CO1
Module – 2					
Q.3	a.	Explain the following with examples : (i) RSC (ii) MLA (iii) STRH (iv) SWP	10	L2	CO2
	b.	Explain the different data processing instruction in ARM.	10	L2	CO2
OR					
Q.4	a.	Explain Barrel shifter instruction in ARM with suitable examples.	10	L2	CO2
	b.	Explain the different branch instruction of ARM processor.	05	L2	CO2
	c.	Explain co-processor instruction of ARM processor.	05	L2	CO2
Module – 3					
Q.5	a.	Explain the different basic data types in C. Provide examples of how each data type can be used in a C program.	08	L2	CO3
	b.	Discuss the concept of register allocation in compiler optimization. Illustrate its significance with an example.	07	L2	CO3
	c.	Describe the process of a function call in C.	05	L2	CO3
OR					
Q.6	a.	Discuss the common portability issues faced when writing C programs. How can these issues be mitigated.	07	L2	CO3
	b.	Explain the concept of pointer aliasing with example.	07	L2	CO3
	c.	How are function calls handled efficiently in calling function in C?	06	L2	CO3
Module – 4					
Q.7	a.	What are interrupts? Discuss interrupt vector table with diagram for ARM processor.	06	L2	CO4
	b.	Describe the sequence of operations that occurs when an ARM processor handles an IRQ exceptions.	06	L2	CO4
	c.	Discuss the priority system for exception in ARM processor.	08	L2	CO4
OR					
Q.8	a.	Explain the role of the link register in ARM exception handling.	08	L2	CO4
	b.	Explain the design and implementation of an interrupt stack in a ARM-based system. Explain the steps involved.	08	L2	CO4
	c.	What are the key differences between a boot loader and firmware?	04	L2	CO4

Module – 5					
Q.9	a.	Explain the basic operation of a cache controller.	06	L2	CO5
	b.	With a neat diagram, explain the basic architecture of a cache memory.	10	L2	CO5
	c.	Mention any 4 relationship between cache and main memory.	04	L2	CO5
OR					
Q.10	a.	Write a note on cache write policy both write back or write through.	10	L2	CO5
	b.	Describe the allocation policy on a cache miss.	04	L2	CO5
	c.	Write a note on following : (i) Write buffers (ii) Cache efficiency	06	L2	CO5

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Fourth Semester B.E. Degree Examination

Subject Title: Microcontrollers

TIME: 03 Hours**Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module -1			*Bloom's Taxonomy Level	Marks
Q.01	a	Mention the difference between 1. Microcontroller and Microprocessor 2. RISC and CISC	L2	10 Marks
	b	Explain the architecture of an ARM embedded device with the help of neat diagram.	L2	10 Marks
OR				
Q.02	a	Explain in detail about Current Program Status Register (CPSR).	L2	10 Marks
	b	With a neat diagram, explain embedded system Hardware.	L2	10 Marks
Module-2				
Q. 03	a	Explain different arithmetic instructions in ARM processor with an example.	L2	10 Marks
	b	Explain single register load store addressing mode syntax, table, index mode with an example.	L2	10 Marks
OR				
Q.04	a	Explain barrel shifter instructions in ARM with suitable examples.	L2	10 Marks
	b	Explain different Logical instructions in ARM processor with an example.	L2	10 Marks
Module-3				
Q. 05	a	Explain code optimization, profiling and cycle counting.	L3	10 Marks
	b	Write a C program that prints the square of the integers between 0 to 9 using functions and explain how to convert this C function to an assembly function with command.	L3	10 Marks
OR				
Q. 06	a	Discuss how registers are allocated to optimize the program.	L3	10 Marks
	b	Develop an ALP to find the sum of first 10 integer numbers.	L3	10 Marks
Module-4				
Q. 07	a	With a neat diagram explain ARM processor exceptions and modes.	L2	10 Marks
	b	Explain assigning interrupts and interrupt latency.	L2	10 Marks
OR				
Q. 08	a	Briefly explain what happens when an IRQ and FIQ exception is raised with an ARM processor.	L2	10 Marks

	b	Explain firmware execution flow and explain Red Hat RedBoot.	L2	10 Marks
Module-5				
Q. 09	a	Explain the basic architecture of cache memory.	L2	10 Marks
	b	Explain how main memory maps to a cache memory.	L2	10 Marks
OR				
Q. 10	a	With a neat block diagram explain associative cache.	L2	10 Marks
	b	Briefly explain cache line replacement policies.	L2	10 Marks

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

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BCS402

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Microcontrollers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Explain the architecture of an arm embedded device with a neat diagram.	10	L2	CO1	
	b.	How are monitor and control internal operations performed in ARM core? Explain in brief.	10	L2	CO1	
OR						
Q.2	a.	Explain memory management in ARM core. Compare cache and tightly coupled memory.	10	L2	CO1	
	b.	Explain mechanism applied by ARM core to handle exception, interrupts using different vector table.	10	L2	CO1	
Module – 2						
Q.3	a.	Examine data processing instructions requirement in the manipulation of data register? Explain in brief data processing instructions.	10	L2	CO2	
	b.	Explain with examples the following 32-bit instruction of ARM processor i) CMN ii) MLA iii) MRS iv) BIC v) LDR.	10	L2	CO2	
OR						
Q.4	a.	Explain the following with example: i) Stack operation ii) Swap instructions.	10	L2	CO2	
	b.	Explain Branch instructions in ARM with suitable example. Demonstrate Branch instruction usage flow of execution with an example program.	10	L2	CO2	
Module – 3						
Q.5	a.	How registers are allocated to optimize the program? Develop an assembly level program to find the sum of first 10 integer numbers.	10	L2	CO3	
	b.	How compiler handles a "for loop" with variable number of iterations N and loop controlling with an example.	10	L2	CO3	
OR						
Q.6	a.	Explain the following terms with an appropriate example: i) Pointer Aliasing ii) Portability issues.	10	L2	CO3	
	b.	How function calling is efficiently used by ARM through APCS with an example program.	10	L2	CO3	
Module – 4						
Q.7	a.	Explain ARM processors exception and modes with a neat diagram.	10	L2	CO4	
	b.	Explain exception priorities and link register offset.	10	L2	CO4	
OR						
Q.8	a.	List ARM firmware suite features. Explain firmware execution flow and Red Hat Boot.	10	L2	CO4	
	b.	Explain IRQ and FIQ exception, also to enable and disable IRQ and FIQ interrupts.	10	L2	CO4	
Module – 5						
Q.9	a.	Explain basic architecture of cache memory.	10	L2	CO5	
	b.	Explain process involved in main memory mapping to a cache memory.	10	L2	CO5	
OR						
Q.10	a.	Explain with diagram set associative cache. How are efficiency is measured?	10	L2	CO5	
	b.	Briefly explain cache line replacement policies with an example.	10	L2	CO5	

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BCS402

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2025 Microcontrollers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the major design rules to implement the RISC design philosophy.	08	L2	CO1
	b.	Differentiate between RISC and CISC processors.	04	L2	CO1
	c.	Explain ARM core data flow model, with neat diagram.	08	L2	CO1
OR					
Q.2	a.	With the help of bit layout diagram, explain Current Program Status Register (CPSR) of ARM.	08	L2	CO1
	b.	With an example, explain the pipeline in ARM.	05	L2	CO1
	c.	Discuss the following with diagrams: (i) Von-Neuman architecture with cache (ii) Harvard architecture with TCM	07	L2	CO1
Module – 2					
Q.3	a.	Explain the different data processing instructions in ARM.	08	L2	CO2
	b.	Explain the different branch instructions of ARM.	04	L2	CO2
	c.	Explain the following ARM instructions: (i) MOV r ₁ , r ₂ (ii) ADDS r ₁ , r ₂ , r ₄ (iii) BIC r ₃ , r ₂ , r ₅ (iv) CMP r ₃ , r ₄ (v) UMLAL r ₁ , r ₂ , r ₃ , r ₄	08	L2	CO2
OR					
Q.4	a.	Explain the different load store instructions in ARM.	08	L2	CO2
	b.	With an example, explain full descending stack operations.	07	L2	CO2
	c.	Develop an ALP to find the sum of first 10 integer numbers.	05	L3	CO2
Module – 3					
Q.5	a.	List out basic C data types used in ARM. Develop a C program to obtain checksums of a data packet containing 64 words and write the compiler output for the above function.	08	L2	CO3
	b.	Explain the C looping structures in ARM.	08	L2	CO3
	c.	Explain pointer aliasing in ARM.	04	L2	CO2

OR

Q.6	a.	With an example, explain function calls in ARM.	08	L2	CO3
	b.	Explain register allocation in ARM.	07	L2	CO3
	c.	Write a brief note on portability issues when porting C code to ARM.	05	L2	CO3

Module – 4

Q.7	a.	Explain the ARM processor exceptions and modes, vector table and exception priorities.	10	L2	CO4
	b.	Explain the interrupts in ARM.	10	L2	CO4

OR

Q.8	a.	Explain the ARM firmware suite and red hat redboot.	10	L2	CO4
	b.	Explain the sandstone directory layout and sandstone code structure.	10	L2	CO4

Module – 5

Q.9	a.	Explain the basic architecture of a cache memory and basic operation of a cache controller.	10	L2	CO5
	b.	With a neat diagram, explain a 4 KB, four way set associative cache.	10	L2	CO5

OR

Q.10	a.	Explain the write buffers and measuring cache efficiency.	08	L2	CO5
	b.	Explain the cache policy.	12	L2	CO5

**Fourth Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024
Microcontrollers**

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	List and explain 4 major design rules in RISC design philosophy.	10	L2	CO1
	b.	With a neat diagram explain ARM based embedded device, a microcontroller.	10	L2	CO1
OR					
Q.2	a.	Explain in detail ARM Design Philosophy.	10	L2	CO1
	b.	Explain in detail software abstraction layers executing on hardware (embedded system software).	10	L2	CO1
Module – 2					
Q.3	a.	Explain arithmetic and logical data processing instructions with syntax, examples and code snippet for each.	10	L3	CO2
	b.	Explain four steps of stack implementation in ARM with examples for each.	10	L2	CO2
OR					
Q.4	a.	Write note on : i) Coprocessor Instructions ii) Software Interrupt instructions	10	L2	CO2
	b.	Develop ARM ALP to find largest number in an array of 32 bit numbers. Program should be neatly commented.	10	L3	CO2
Module – 3					
Q.5	a.	Discuss with C function and target ARM assembly code how optimization can be done with respect to data types.	10	L2	CO3
	b.	Explain optimizations with respect to C loop structures considering fixed number of iterations, variable number of iterations and loop unrolling. Explain with examples for each.	10	L2	CO3
OR					
Q.6	a.	Discuss and analyze optimization with respect to pointer aliasing, effects of pointer aliasing with simple C code / function.	10	L3	CO3
	b.	Analyze and explain the way structure arrangement to be done in order to access the structure members efficiently.	10	L3	CO3
Module – 4					
Q.7	a.	Discuss the following : i) Interrupt latency ii) Types of interrupts available on ARM processor.	10	L2	CO4
	b.	Write short code snippet to enable and disable interrupts. Explain in detail.	10	L3	CO4
OR					
Q.8	a.	What is firmware? Explain firmware execution flow and bootloader.	10	L2	CO4
	b.	Explain / Discuss Sandstone execution flow in detail.	10	L2	CO4
Module – 5					
Q.9	a.	Discuss basic architecture of a cache memory with a neat diagram.	10	L2	CO5
	b.	Explain in detail memory hierarchy and cache memory.	10	L2	CO5
OR					
Q.10	a.	Discuss Cache policy in detail.	10	L2	CO5

MODULE-1

Passing Module

- 1. Explain the architecture of an arm embedded device with a neat diagram.**
- 2. How are monitor and control internal operations performed in ARM core? Explain in brief.**
- 3. Explain memory management in ARM core. Compare cache and tightly coupled memory.**
- 4. Explain mechanism applied by ARM core to handle exception, interrupts using different vector table.**
- 5. Mention the difference between Microcontroller and Microprocessor RISC and CISC**
- 6. Explain in detail about Current Program Status Register (CPSR).**
- 7. With a neat diagram, explain embedded system Hardware.**
- 8. Explain the ARM design philosophy. Explain the core extensions of ARM processor with neat block diagram.**
- 9. What is pipelines in ARM? Illustrate with an example the pipeline stage ARM 9 and ARM 10.**

MODULE- 4

Passing Module

Explain ARM processors exception and modes with a neat diagram.

Explain exception priorities and link register offset.

List ARM firmware suite features. Explain firmware execution flow and Red Hat Boot.

Explain IRQ and Fir exception, also to enable and disable IRQ and FIQ interrupts.

Explain assigning interrupts and interrupt latency.

What are interrupts? Discuss interrupt vector table with diagram for ARM processor.

Describe the sequence of operations that occurs when an ARM processor handles an IRQ exceptions.

**Explain the design and implementation of an interrupt stack in a ARM-based system.
Explain the steps involved.**

What are the key differences between a boot loader and firmware?

MODULE- 5

Passing Module

Explain basic architecture of cache memory.

Explain process involved in main memory mapping to a cache memory.

Explain with diagram set associative cache. How are efficiency is measured?

Briefly explain cache line replacement policies with an example.

Explain the basic operation of a cache controller.

Mention any 4 relationship between cache and main memory.

Write a note on cache write policy both write back or write through.

Describe the allocation policy on a cache miss.

Write a note on following:

(i) Write buffers (ii) Cache efficiency

MODULE- 3

Explain the different basic data types in C. Provide examples of how each data type can be used in a C program.

Discuss the concept of register allocation in compiler optimization. Illustrate its significance with an example.

Describe the process of a function call in C.

Discuss the common portability issues faced when writing C programs. How can these issues be mitigated.

Explain the concept of pointer aliasing and portability issues with example.

How are function calls handled efficiently in calling function in C?

Explain code optimization, profiling and cycle counting.

Develop an ALP to find the sum of first 10 integer numbers.

How registers are allocated to optimize the program? Develop an assembly level program to find the sum of first 10 integer numbers.

How compiler handles a "for loop" with variable number of iterations N and loop controlling with an example.

How function calling is efficiently used by ARM through APCS with an example program.

MODULE- 2

Examine data processing instructions requirement in the manipulation of data register? Explain in brief data processing instructions.

Explain with examples the following 32-bit instruction of ARM processor

- i) MLA**
- ii) MRS**
- iii) BIG.**
- iv) LDR.**

Explain the following with example:

- i) Stock operation ii) Swap Instructions.**

Explain Branch instructions in ARM with suitable example. Demonstrate Branch instruct usage flow of execution with an example program.

Explain different Logical instructions in ARM processor with an example.

Explain single register load store addressing mode syntax, table, index mode with an example.

Explain barrel shifter instructions in ARM with suitable examples & Explain co-processor instruction of ARM processor.

Explain different arithmetic instructions in ARM processor with an example.